

INDUSTRIAL TECHNOLOGY

Overview

The Department of Agricultural and Biosystems Engineering offers a Bachelor of Science degree in Industrial Technology (ITEC), as well as an undergraduate certificate in Occupational Safety. Students majoring in ITEC choose between three options: Biological and Biomanufacturing Systems Technology, Manufacturing or Occupational Safety. The department also offers a minor in Industrial Technology.

Successful ITEC graduates gain knowledge, skills, and abilities in solving technical problems, understanding the design process, excelling in authentic leadership, being aware of safety issues, having a quality orientation, effectively managing projects, and having a systems-thinking perspective. This translates to a holistic approach that uses science and engineering principles to focus on the way the constituent parts of a manufacturing system interrelate, how they work over time, and how they fit the context of larger systems. Graduates find careers within a variety of industries, businesses, and organizations in the fields of advanced manufacturing and biomanufacturing; robotics; analytical and processing laboratories; automation and controls; biofuel and beverage fermentation systems electronics; lean manufacturing; quality and operations management; safety management, loss prevention; or industrial hygiene.

Common job duties of ITEC Biological and Biomanufacturing Systems Technology graduates include:

- Management of technical projects
- Implementation, management, and maintenance of instruments and equipment
- Management of bioprocessing and biotechnology systems

Common job duties of ITEC Manufacturing graduates include:

- Supervision of production systems
- Design of manufacturing processes and products

Common job duties of ITEC Occupational Safety graduates include:

- Development, management, and evaluation of safety programs and systems
- Hazard identification and mitigation
- Loss prevention

The certificate in occupational safety is designed to meet the needs of the students who will find themselves in management roles with responsibilities that include safety. The certificate program prepares technically oriented managers to meet their professional safety responsibilities

For more information about the occupational safety certificate: <http://www.abe.iastate.edu/home/certificate-in-occupational-safety/>

Student learning outcomes

Upon graduation, all ITEC students should be able to:

1. Apply knowledge of mathematics, science, computation, and applied engineering to identify and solve applied science and technology problems.
2. Develop and conduct experiments and analyze and interpret resulting data.
3. Evaluate and adapt systems, processes and programs to meet desired needs.
4. Function effectively on multi-disciplinary teams
5. Communicate effectively, ethically, and professionally in written, oral, and other formats to technical and non-technical audiences.
6. Understand the potential impacts and limitations of solutions in global and societal contexts.
7. Recognize the need for, and demonstrate an ability to, engage in life-long learning.
8. Effectively apply modern scientific and technical tools necessary for professional practice to address contemporary issues in applied engineering and technology.

Upon graduation, ITEC students in the biological and biomanufacturing systems technology (BBST) option should be able to:

1. Develop, implement, troubleshoot, and evaluate bio-based manufacturing processes, facilities, and products.
2. Calculate and interpret mass and energy balances of biological processes involving microbial growth and conversion, plants, animals, and humans to optimize processing and production systems.
3. Calculate and interpret rates of biological processes involving reaction kinetics, growth, death, heat production, and transpiration to optimize processing and production systems.

Upon graduation, ITEC students in the manufacturing (M) option should be able to:

1. Create, implement, and evaluate manufacturing processes and facility plans.
2. Integrate and apply tools in computer aided design, manufacturing, controls, robotics, and automation systems to applied engineering and technology management settings.
3. Evaluate technologies to enhance production, quality, sustainability, and profitability of manufacturing systems and facility management.

Upon graduation, ITEC students in the occupational safety (OS) option should be able to:

1. Design, implement, and evaluate occupational safety and health programs for work environments.
2. Identify, assess, and analyze hazards and loss-producing conditions in work environments.
3. Eliminate or control occupational hazards using appropriate technologies, training, and administrative interventions.

ACCREDITATION

For the undergraduate curriculum in Industrial Technology leading to a Bachelor of Science. The Industrial Technology program is accredited by the Association of Technology, Management, and Applied Engineering, using standards to accredit baccalaureate programs, <https://atmae.org/accreditation/>. ATMAE accredits degree programs in technology, technology management, applied technology, engineering technology, applied engineering, and technology-related disciplines delivered at nationally and regionally accredited institutions in the United States and institutions meeting the same standards internationally.

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For more information about the ITEC degree: <https://www.abe.iastate.edu/undergraduate-students/industrial-technology/>

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- **Department Website** – Industrial Technology – Department of Agricultural and Biosystems Engineering (<https://www.abe.iastate.edu/undergraduate-students/industrial-technology/>)

Program Administration

- **Owning Academic Unit/College** – College of Agriculture and Life Sciences
- **Coordinating Academic Unit Department** – Agricultural and Biosystems Engineering - A Department

credential opportunities

- **Program Duration** – 4 years
- **Credential(s) Offered**
 - Industrial Technology, B.S. (120 credits)
- **Program Focus Areas:**
 - Biological and Biomanufacturing Systems Technology Undergraduate Program Focus

- Manufacturing Undergraduate Program Focus
- Occupational Safety Undergraduate Program Focus

• **Additional Major Note** – Industrial Technology is offered as an additional major.

• **Program Limitations** – None

• **Program Designation** – In-person

Total Degree Requirement: 120 cr.

Only 65 cr. from a two-year institution may apply which may include up to 16 technical cr.; 9 P-NP cr. of free electives; 2.00 minimum GPA.

Communications Proficiency:

6 cr. of English composition with a C or better and 3 cr. of speech fundamentals with a C or better.

Communication/Library: 13 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
LIB 1600	Introduction to College Level Research	1
One of the following:		3
ENGL 3020	Business Communication	
ENGL 3090	Proposal and Report Writing	
ENGL 3140	Technical Communication	
AGEDS 3270	Survey of Agriculture and Life Sciences Communication	
One of the following:		3
SPCM 2120	Fundamentals of Public Speaking	
COMST 2140	Professional Communication	
AGEDS 3110	Presentation and Sales Strategies for Agricultural Audiences	
Total Credits		13

Mathematical, Physical, and Life Sciences: 25 cr.

STAT 1040	Introduction to Statistics	3
MATH 1450	Applied Trigonometry	3
MATH 1510	Calculus for Business and Social Sciences	3
PHYS 1310	General Physics I	4
PHYS 1310L	General Physics I Laboratory	1
CHEM 1630	College Chemistry	4
CHEM 1630L	Laboratory in College Chemistry	1
One of the following:		3
BIOL 1010	Introductory Biology	
BIOL 2110	Principles of Biology I	
BIOL 2120	Principles of Biology II	
BIOL 2510	Biological Processes in the Environment	

Second Biology course requirement by Option:	3
Biological & Biomanufacturing Systems Technology options	
BIOL 2120	
Manufacturing options	
Life Sciences Elective from approved College of Agriculture and Life Sciences list	
Occupational Safety option	
BIOL 2550 Fundamentals of Human Anatomy	
Total Credits	25

Business, Humanities, Ethics, and Social Sciences: 18 cr.

Occupational safety option must take ACCT 2150

ACCT 2840 Financial Accounting	3
or ACCT 2150 Legal Environment of Business	
ECON 1010 Principles of Microeconomics	3
TSM 3700 Occupational Safety (Ethics)	3
Humanities course from College of Agriculture and Life Sciences list	3
International Perspectives course from University list	3
U.S. Cultures and Communities course from University list	3
Total Credits	18

Technical Core: 27 cr.

TSM 1100 Introduction to Technology	1
TSM 1110 Experiencing Technology	1
TSM 1150 Solving Technology Problems	3
TSM 1160 Introduction to Design in Technology	3
TSM 2010 Preparing for Workplace Seminar	1
TSM 2100 Fundamentals of Technology	3
TSM 2140 Managing Technology Projects	1
TSM 3100 Total Quality Improvement	3
TSM 3631x: Electrical Power, Control, and Sensor Systems	4
TSM 3710 Occupational Safety Management	2
TSM 3970 Summer Internship in Technology	R
or TSM 3990 Internship in Technology	
TSM 4150 Applied Project Management in Technology	2
TSM 4160 Technology Capstone	3
Total Credits	27

TSM 3970 or TSM 3990 may count toward graduation.**Biological and Biomanufacturing Systems Technology Option: 37 cr.**

BIOL 2120 Principles of Biology II	3
CHEM 2110 Quantitative and Environmental Analysis	2
CHEM 2110L Quantitative and Environmental Analysis Laboratory	2

LDST 3220 Leadership Practice	3
MICRO 2010 Introduction to Microbiology	2
or MICRO 3020 Biology of Microorganisms	
MICRO 3000L Introductory Microbiology Laboratory	1
TSM 3800 Fundamentals, Applications, and Modeling of Biological Systems	3
Select 15 credits from departmental list	15
6 credits of free electives	6
Total Credits	37

Manufacturing Option: 37 cr.

One of the following:	1
ABE 2710 Engineering Applications of Parametric Solid Modeling	
ABE 2720 Parametric Solid Models, Drawings, and Assemblies Using Creo Parametric	
ABE 2730 CAD for Process Facilities and Land Use Planning	
TSM 2160 Advanced Technical Graphics, Interpretation, and CAD	2
TSM 2400 Introduction to Advanced Manufacturing: Metals and Plastics Processing	3
TSM 2415 Introduction to Lean Manufacturing Systems	3
TSM 3370 Fluid Power Systems Technology	3
TSM 3400 Advanced Automated Manufacturing Processes	3
TSM 3650 Electrical Controls and Sensor Systems	3
TSM 4430 Statics and Strength of Materials for Technology	3
TSM 4440 Facility Planning	3
TSM 4650 Automation Systems	3
10 credits of free electives	10
Total Credits	37

Occupational Safety Option: 37 cr.

TSM 2400 Introduction to Advanced Manufacturing: Metals and Plastics Processing	3
TSM 3720 Legal Aspects of Occupational Safety and Health	2
TSM 3760 Fire Protection and Prevention	3
TSM 4700 Basics of Industrial Hygiene	3
TSM 4710 Safety Laboratory	1
TSM 4770 Risk Analysis and Management	3
HS 1050 First Aid and Emergency Care	2
PSYCH 2500 Psychology of the Workplace	3
17 credits of free electives	17
Total Credits	37

Industrial Technology, B.S. - Manufacturing option**First Year**

Fall	Credits	Spring	Credits
TSM 1100		1 TSM 1110	1
TSM 1160		3 TSM 1150	3
ENGL 1500		3 MATH 1510	3
LIB 1600		1 PHYS 1310	4
MATH 1450		3 PHYS 1310L	1
CHEM 1630		4 TSM 2400	3
CHEM 1630L		1	
16		15	

Second Year

Fall	Credits	Spring	Credits
TSM 2010		1 TSM 2160	2
TSM 2140		1 TSM 2415	3
TSM 2100		3 STAT 1040	3
TSM 3700		3 BIOL 1010 or 2110	3
ECON 1010		3 International Perspectives - See list*	3
ENGL 2500		3 SPCM 2120, COMST 2140, or AGEDS 3110	3
ABE 2710, 2720, or 2730		1	
15		17	

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
TSM 3100		3 TSM 3710		2 TSM 3970 or 3990	R
TSM 3370		3 TSM 3650		3	
TSM 3631x		4 ACCT 2840 or 2150		3	
U.S. Cultures and Communities - See list*		3 ENGL 3020, 3090, 3140, or AGEDS 3270		3	

Humanities - See list	3 Life Science - See list*	3
16		14
		0

Fourth Year

Fall	Credits	Spring	Credits
TSM 4150		2 TSM 4160	3
TSM 3400		3 TSM 4440	3
TSM 4430		3 TSM 4650	3
Elective		5 Elective	5
13		14	

* International Perspectives course list (<https://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current/>)
 U.S. Cultures and Communities course list (<https://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses/>)
 Humanities course list (<https://www.cals.iastate.edu/student-services/humanities/>)
 Life Science course list (<https://www.cals.iastate.edu/student-services/life-science/>)

Industrial Technology, B.S. - Occupational Safety option**First Year**

Fall	Credits	Spring	Credits
TSM 1100		1 TSM 1110	1
TSM 1160		3 TSM 1150	3
ENGL 1500		3 MATH 1510	3
LIB 1600		1 PHYS 1310	4
MATH 1450		3 PHYS 1310L	1
CHEM 1630		4 ECON 1010	3
CHEM 1630L		1	
16		15	

Second Year

Fall	Credits	Spring	Credits
TSM 2010		1 TSM 2400	3
TSM 2140		1 TSM 3710	2
TSM 2100		3 HS 1050	2
TSM 3700		3 STAT 1040	3
ENGL 2500		3 SPCM 2120, COMST 2140, or AGEDS 3110	3

BIOL 2510	3 BIOL 2550	3
14	16	

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
TSM 3720		2 TSM 3100		3 TSM 3970 or 3990	R
TSM 3631x		4 TSM 4710		1	
TSM 3760		3 TSM 4700		3	
ENGL 3020, 3090, 3140, or AGEDS 3270		3 ACCT 2150		3	
Elective		4 International Perspectives *		3	
16		13		0	

Fourth Year

Fall	Credits	Spring	Credits
TSM 4150		2 TSM 4160	3
TSM 4770		3 U.S. Cultures and Communities*	3
PSYCH 2500		3 Humanities*	3
Elective		7 Elective	6
15		15	

* International Perspectives course list (<https://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current/>)
 U.S. Cultures and Communities course list (<https://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses/>)
 Humanities course list (<https://www.cals.iastate.edu/student-services/humanities/>)
 Life Science course list (<https://www.cals.iastate.edu/student-services/life-science/>)

Industrial Technology, B.S. - Biological and Biomanufacturing Systems Technology option

First Year

Fall	Credits	Spring	Credits
TSM 1100		1 TSM 1110	1
TSM 1160		3 TSM 1150	3
ENGL 1500		3 MATH 1510	3
LIB 1600		1 PHYS 1310	4
MATH 1450		3 PHYS 1310L	1

CHEM 1630	4 ECON 1010	3
CHEM 1630L	1	
16	15	

Second Year

Fall	Credits	Spring	Credits
TSM 2010		1 TSM 3700	3
TSM 2140		1 BIOL 2120	3
TSM 2100		3 CHEM 2110	2
BIOL 2110		3 CHEM 2110L	2
ENGL 2500		3 LDST 3220 (US Cultures & Communities)	3
Option core course		3 STAT 1040	3
14		16	

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
TSM 3100		3 TSM 3631x		4 TSM 3970 or 3990	R
MICRO 2010		2 TSM 3710		2	
MICRO 3000L		1 TSM 3800		3	
ACCT 2840 or 2150		3 Humanities - See list *		3	
Speech Communication - See list *		3 International Perspectives - See list *		3	
Option core course - See list *		3			
15		15		0	

Fourth Year

Fall	Credits	Spring	Credits
TSM 4150		2 TSM 4160	3
Option core course - See list *		3 Advanced English - See list *	3
Option core course - See list *		3 Option core course - See list *	3
Elective		6 Elective	6
14		15	

- * International Perspectives course list (<https://www.registrar.iastate.edu/students/div-ip-guide/IntlPerspectives-current/>)
- U.S. Cultures and Communities course list (<https://www.registrar.iastate.edu/students/div-ip-guide/usdiversity-courses/>)
- Humanities course list (<https://www.cals.iastate.edu/student-services/humanities/>)
- Life Science course list (<https://www.cals.iastate.edu/student-services/life-science/>)
- Option Core course list (<https://www.abe.iastate.edu/undergraduate-students/industrial-technology/itec-curricula/>)

Minor - Industrial Technology

The Department of Agricultural and Biosystems Engineering offers a minor in industrial technology which may be earned by completing a minimum of 15 credits of technology systems management (TSM) courses, which includes:

TSM 1150	Solving Technology Problems	3
TSM 2100	Fundamentals of Technology	3
9 credits from:		9
TSM 2160	Advanced Technical Graphics, Interpretation, and CAD	
TSM 2400	Introduction to Advanced Manufacturing: Metals and Plastics Processing	
TSM 2415	Introduction to Lean Manufacturing Systems	
TSM 3100	Total Quality Improvement	
TSM 3370	Fluid Power Systems Technology	
TSM 3400	Advanced Automated Manufacturing Processes	
TSM 3631x	Electrical Power, Control, and Sensor Systems	
TSM 3650	Electrical Controls and Sensor Systems	
TSM 3700	Occupational Safety	
TSM 3710	Occupational Safety Management	
TSM 3720	Legal Aspects of Occupational Safety and Health	
TSM 3760	Fire Protection and Prevention	
TSM 4430	Statics and Strength of Materials for Technology	
TSM 4415	Lean Manufacturing Systems	
TSM 4440	Facility Planning	
TSM 4650	Automation Systems	
TSM 4700	Basics of Industrial Hygiene	
TSM 4710	Safety Laboratory	
TSM 4770	Risk Analysis and Management	

- At least six (6) credits of 3000-level or higher TSM classes (from the courses listed above)

- At least three (3) credits that are not used to meet any other department, college, or university requirement.

Total Credits

15

For the undergraduate curriculum in agricultural systems technology leading to the degree of Bachelor of Science or for the undergraduate curriculum in industrial technology leading to the degree of Bachelor of Science.

The department also offers an undergraduate curricula and courses in agricultural engineering, biological systems engineering.

Certificate in Occupational Safety

The Department of Agricultural and Biosystems Engineering offers an undergraduate certificate in occupational safety (<http://www.abe.iastate.edu/undergraduate-students/industrial-technology/certificate-in-occupational-safety/>) which may be earned by completing a minimum of 20 credits of technology systems management courses, which includes:

TSM 3700	Occupational Safety	3
TSM 3710	Occupational Safety Management	2
TSM 3720	Legal Aspects of Occupational Safety and Health	2
TSM 4700	Basics of Industrial Hygiene	3
6 credits from a departmentally approved list		9
TSM 4930D	Workshop in Technology: Occupational Safety (Note: This course needs to be the last course taken toward completion of the Occupational Safety Certificate)	1-4